

CHAPTER 2

SYSTEM ANALYSIS

2.1 Introduction

System analysis is a critical phase in the development of any software system, as it focuses on understanding the problem domain, identifying user requirements, and defining the functional and non-functional aspects of the proposed solution. For healthcare-oriented applications such as MedSure, system analysis plays an even more significant role due to the sensitive nature of medical data, the need for accuracy, and the requirement for strong security and privacy controls. This phase ensures that the system is designed to meet real-world healthcare needs while complying with data protection standards and ethical considerations.

MedSure is designed as a secure, intelligent, and role-based medical report and prescription analysis platform. The system serves multiple user groups including patients, administrators, and medical stores, each with distinct responsibilities and access levels. During system analysis, the workflows of these users are carefully examined to ensure smooth interaction, minimal complexity, and maximum reliability. The analysis phase also evaluates how users upload prescriptions and medical reports, how data is securely stored, and how artificial intelligence is safely integrated to generate meaningful medical insights without exposing personal patient information.

This chapter presents a detailed analysis of the existing challenges in traditional prescription handling systems and explains how MedSure addresses these limitations through a modern, technology-driven approach. It discusses system requirements, user roles, data flow, security constraints, and AI integration strategies. Special emphasis is placed on protecting patient privacy by sanitizing sensitive information before AI processing. By thoroughly analyzing both user expectations and system constraints, this chapter lays a strong foundation for the design and implementation of a reliable, scalable, and secure medical analysis platform.

2.2 Existing System

The existing system for handling medical prescriptions and hospital reports is largely manual, fragmented, and inefficient. In most healthcare environments, prescriptions are handwritten by doctors or provided as printed documents, while medical reports are issued as physical papers or PDF files. Patients are expected to preserve these documents on their own, often storing them in files or folders without any structured organization. Over time, prescriptions and reports may be lost, damaged, or forgotten, making it difficult for patients to track their medical history accurately.

In the current system, understanding prescriptions heavily depends on direct interaction with doctors or pharmacists. Medical terminology, dosage instructions, and drug interactions are often complex and not easily understood by patients. Many patients rely on verbal explanations, which may be unclear or forgotten later. There is no centralized digital platform that allows patients to upload prescriptions and receive structured, simplified explanations or warnings. As a result, patients may misuse medicines, miss dosage schedules, or consume duplicate or conflicting drugs, which can lead to serious health risks.

Another major limitation of the existing system is the lack of medicine verification and availability tracking. Patients usually trust the medical store or supplier without having a way to verify whether a medicine is genuine or counterfeit. Information about medicine authenticity, manufacturer details, side effects, and nearby availability is scattered across multiple sources and often unreliable. Additionally, medical data is rarely digitized in a secure and searchable format, making it difficult for healthcare providers or patients to retrieve historical records when needed.

From a technological perspective, most existing solutions do not leverage artificial intelligence effectively. Even where digital health platforms exist, they typically provide basic record storage or appointment booking features without intelligent analysis of prescriptions or reports. There is little to no automated extraction of text from medical documents, no contextual understanding of medical data, and no privacy-aware AI processing. Furthermore, data security is often weak, with patient information stored without proper encryption or access control mechanisms.

Overall, the existing system lacks automation, intelligence, personalization, and security. It places a heavy burden on patients to manage their own medical records and understand complex medical information without adequate technological support. These limitations highlight the need for a secure, AI-powered, and user-friendly platform like MedSure, which aims to modernize prescription analysis, enhance patient awareness, and protect sensitive healthcare data.

2.2.1 Limitations of Existing System

- Manual handling of handwritten prescriptions and paper-based medical records
- Difficulty for patients to understand medical terms and dosage instructions
- No mechanism to verify the authenticity of prescribed medicines
- Absence of automated prescription analysis and intelligent recommendations
- Inadequate data security, privacy, and access control mechanisms

2.3 Proposed System

The proposed system, **MedSure**, is a secure and intelligent healthcare web application developed to modernize the way medical prescriptions and reports are managed, analyzed, and understood. The system is designed to help patients easily upload their prescriptions and medical documents and receive clear, meaningful explanations and suggestions. By digitizing medical

records and automating analysis, MedSure reduces dependency on manual interpretation and minimizes errors caused by unreadable handwriting or lack of medical knowledge.

MedSure follows a well-structured workflow that ensures both accuracy and privacy. Once a patient uploads a prescription or report, the file is stored securely in private cloud storage. The backend system applies Optical Character Recognition (OCR) to extract text from the document. This extracted content is then sanitized using data anonymization techniques to remove or mask sensitive patient information before any further processing. This ensures that personal and confidential data remains protected at all stages of the system.

After sanitization, the cleaned medical text is analyzed using a Large Language Model (LLM). The AI engine understands the medical context and generates structured outputs such as simplified prescription summaries, medication usage instructions, possible side effects, dosage warnings, and general health guidance. The results are presented in a user-friendly format that is easy for patients to understand, helping them make informed decisions about their treatment and follow medical instructions more accurately.

In addition to patient-focused features, MedSure includes an administrative module that supports system management and oversight. Administrators can manage users, review system logs, monitor prescription processing activity, and ensure compliance with healthcare data protection standards. The system is scalable and modular, making it suitable for hospitals, clinics, pharmacies, and telemedicine platforms. By integrating intelligent automation with strong security controls, MedSure provides a reliable and future-ready solution for digital healthcare management.

2.3.1 Advantages of the Proposed System

- Ensures **secure handling of sensitive patient data** through encryption, access control, and data anonymization techniques.
- Provides **accurate and easy-to-understand analysis** of prescriptions and medical reports using AI-based processing.
- Reduces **manual effort and human errors** caused by illegible handwriting or misinterpretation of medical documents.
- Offers **role-based access** for patients, administrators, and medical stores, ensuring proper data control and authorization.
- Enables **faster decision-making** by delivering real-time summaries, warnings, and medication suggestions.
- Improves **patient awareness and medication compliance** by presenting clear dosage instructions and precautions.
- Designed to be **scalable and adaptable**, allowing integration with hospitals, clinics, pharmacies, and future healthcare services.

2.4 Feasibility Study

A feasibility study evaluates whether the proposed **MedSure – AI-Powered Prescription and Medical Document Analysis & Medicine Validation System** can be successfully developed and implemented within the given constraints. This study analyzes the project from technical, economic, operational, and time perspectives to ensure practicality and effectiveness.

The project is technically feasible as it uses modern and widely adopted technologies such as **Next.js, NestJS, Node.js, MongoDB, OCR tools, and AI/LLM models**. These technologies are well-documented, open-source, and supported by strong developer communities. Cloud services like AWS (for private storage) and existing OCR libraries make it possible to process prescriptions and reports efficiently. The modular architecture ensures smooth integration between frontend, backend, database, and AI components.

MedSure is economically feasible because it relies mostly on **open-source frameworks and libraries**, reducing development costs. Initial deployment can be done using free or low-cost cloud tiers. Since this is an academic project, operational costs remain minimal while still delivering a production-like system. In real-world use, the system can be scaled based on demand, making it cost-effective for healthcare institutions.

The system is user-friendly and designed for **patients, administrators, and medical stores**, each having a dedicated and simple interface. Patients can upload and manage their medical documents easily, while administrators can oversee system operations securely. Minimal training is required as the workflows closely align with real-world healthcare processes. Strong security

The project is feasible within the given **1.5-month timeframe** by following a phased development approach. Core features such as user authentication, document upload, OCR processing, and basic AI analysis can be implemented first, followed by enhancements. Using well-structured frameworks like NestJS and Next.js helps accelerate development and maintain code quality within the limited time.

2.5 Software Engineering Paradigm Applied

The development of **MedSure – AI-Powered Prescription and Medical Document Analysis & Medicine Validation System** follows the **Agile Software Engineering Paradigm**. Agile methodology is chosen because it supports iterative development, flexibility, and continuous improvement, which are essential for a healthcare-based application involving evolving requirements, security concerns, and AI integration.

Under the Agile paradigm, the system is developed in **small, manageable iterations (sprints)**. Each sprint focuses on a specific module such as user authentication, prescription upload, OCR processing, AI-based analysis, or admin management. This approach allows early detection of issues, continuous testing, and regular feedback, ensuring that the system evolves in alignment with user needs and project goals.

Agile also enables **incremental integration of advanced features** like AI-driven suggestions, medicine validation, and privacy-preserving data handling. Core functionalities are implemented first, and intelligent components are enhanced over time without disrupting the existing system. This reduces development risk and improves overall system reliability.

Additionally, Agile promotes **collaboration and adaptability**, making it suitable for a project that combines frontend development, backend services, database management, and AI processing. By adopting the Agile paradigm, MedSure ensures faster development, improved quality, and the ability to respond effectively to changes, resulting in a robust and user-centric healthcare application.